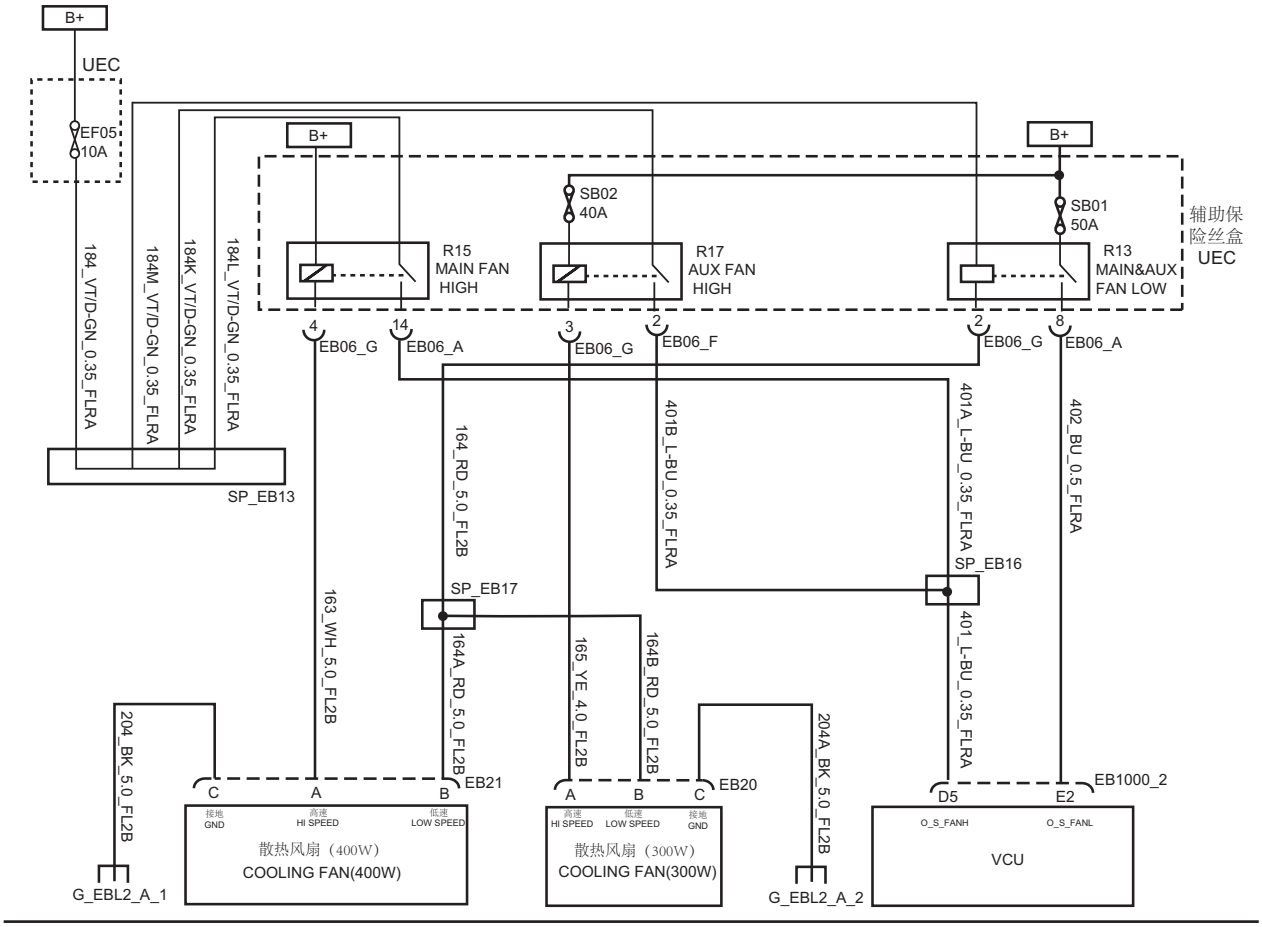
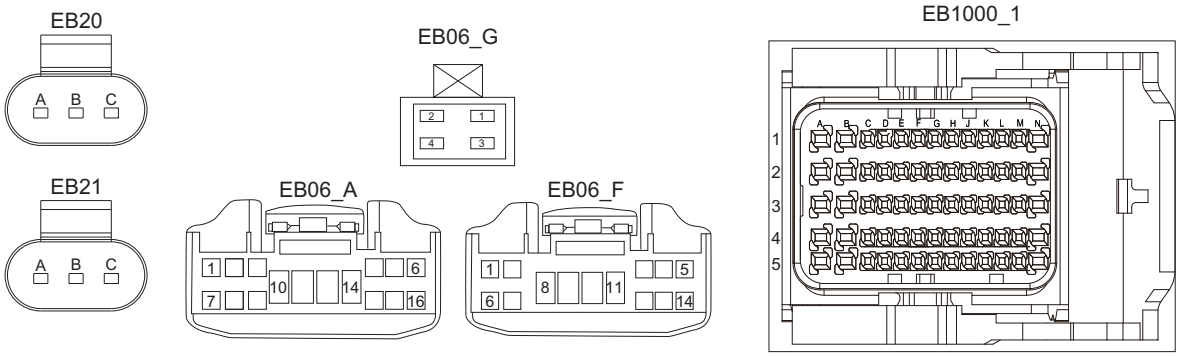
P0692、 P0691、 P0480、 P0694、 P0693、 P0481

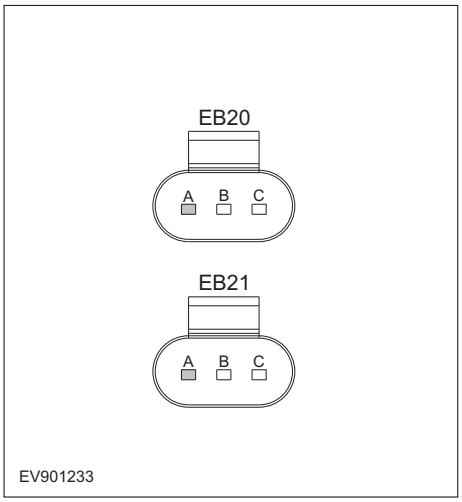
Fault diagnosis code
P0692：DEM_stFanRly1SBCfmd
P0691：DEM_stFanRly1SGCfmd
P0480：DEM_stFanRly1OCCfmd
P0694：DEM_stFanRly2SBCfmd
P0693：DEM_stFanRly2SGCfmd
P0481：DEM_stFanRly2OCCfmd
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • VCU module fault • Cooling fan relay fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0692
The underlying diagnosis
P0691
The underlying diagnosis
P0480
The underlying diagnosis
P0694
The underlying diagnosis
P0693
The underlying diagnosis
P0481
The underlying diagnosis
To set the effect of a fault code condition
Description of circuit diagram

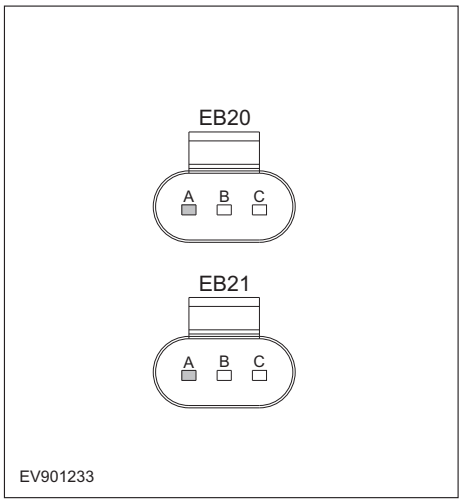
Circuit diagram

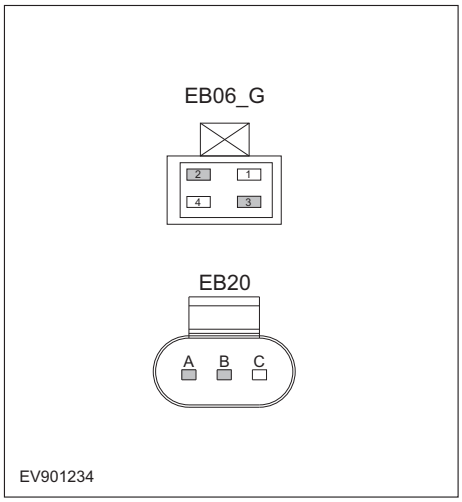
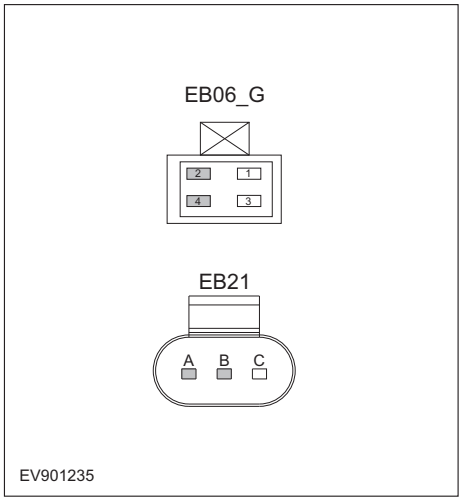




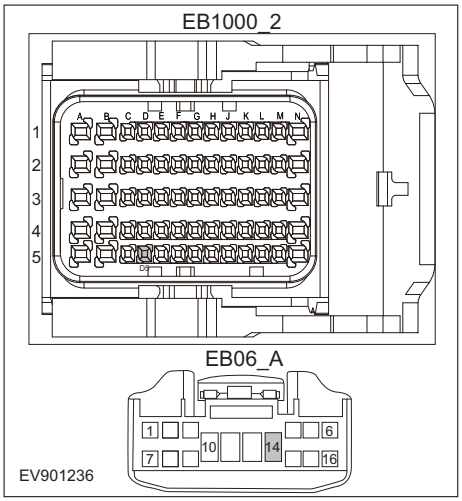
EV901210

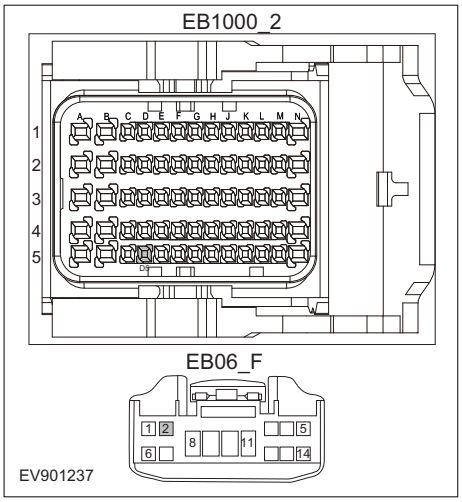
<table>
<tr><td colspan="3">Troubleshooting steps</td></tr>
<tr><td>Step 1</td><td colspan="2">General inspection.</td></tr>
<tr><td colspan="2"></td><td>1 Check fuses SB01/SB02/EF05 for damage.
2 Check the relay R13, R15/R17 for damage.
3 Check battery capacity.
4 Check the harness plug of cooling fan , VCU for damage, poor contact, aging and looseness.
5 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO → Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
<tr><td>Step 2</td><td colspan="2">Check fault codes.</td></tr>
<tr><td colspan="2"></td><td>1 Connect the fault diagnosis instrument to the diagnosis interface.
2 Turn the ignition switch to ON position.
3 Diagnose the VCU with diagnostic apparatus.
4 If there is any fault code except for P0692、P0691、P0480、P0694、P0693、P0481?</td></tr>
<tr><td colspan="2"></td><td>YES → Refer to:DTC Summary list(VCU).</td></tr>
<tr><td colspan="3">NO ↓</td></tr>
<tr><td>Step 3</td><td colspan="2">Check the power voltage of cooling fan battery.</td></tr>
<tr><td colspan="2"> 
</td><td>1 Turn the ignition switch to OFF.
2 Disconnect the harness plug EB20, EB21 of cooling fan.
3 Turn the ignition switch to ON.
4 Measure the voltage between the harness terminal of cooling fan and ground.
Measuring circuit: voltage between terminal A on plug EB20 and ground.
Measuring circuit: voltage between terminal A on plug EB21 and ground.
Standard value: 0V
5 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO → Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
</table>

Step 4	Check the power voltage when the cooling fan is started or operating
	1 Turn the ignition switch to ON. 2 Step down the brake pedal and make the cooling fan relay in working state. 3 Measure the voltage between the harness terminal of cooling fan and ground. Measuring circuit: voltage between terminal A on plug EB20 and ground. Measuring circuit: voltage between terminal A on plug EB21 and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the circuit between cooling fan relay and cooling fan (open circuit).
 	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB06_G of UEC. 3 Measure the resistance between the harness terminal of UEC and the cooling fan. Measuring circuit: resistance between terminal 3 on plug EB06_G and terminal A on plug EB20. Measuring circuit: resistance between terminal 2 on plug EB06_G and terminal B on plug EB20. Measuring circuit: resistance between terminal 4 on plug EB06_G and terminal A on plug EB21. Measuring circuit: resistance between terminal 2 on plug EB06_G and terminal B on plug EB21. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the circuit between cooling fan relay and VCU (open circuit).





1 Turn the ignition switch to OFF.

2 Disconnect the harness plug EB06_A, EB06_F of UEC module.

3 Disconnect the harness plug EB1000_2 of GAW module.

4 Measure the resistance between the harness terminal of UEC and the VCU.

Measuring circuit: resistance between terminal 14 on plug EB06_A and terminal D5 on plug EB1000_2.

Measuring circuit: resistance between terminal 2 on plug EB06_F and terminal D5 on plug EB1000_2.

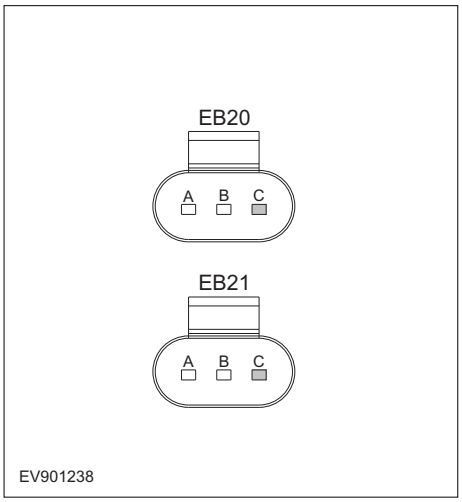
Standard value: ≈ 0Ω

5 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 7	Check the grounding circuit of the cooling fan.



1 Turn the ignition switch to OFF.

2 Measure the resistance between the harness terminal of cooling fan and ground.

Measuring circuit: resistance between terminal C on plug EB20 and ground.

Measuring circuit: resistance between terminal C on plug EB21 and ground.

Standard value: ≈ 0Ω

3 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 8	Check the relay of cooling fan. (components)
	1 Turn the ignition switch to OFF. 2 Replace the relay of cooling fan in good condition. 3 Connect the fault diagnosis instrument to the diagnosis interface. 4 Turn the ignition switch to ON position. 5 Clear fault codes. 6 Reread fault codes. 7 Test whether the faulted code is existed.
	NO → Replace the relay of cooling fan.
YES ↓	
Step 9	Test whether VCU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1000_1, IP55_A of VCU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the VCU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	